Hippolyte Mayard

hippolytemayard@gmail.com

GitHub LinkedIn Google Scholar Personal webpage

Senior ML Scientist - Medical Imaging & Computer Vision | Research to Production

6+ years shipping computer-vision models to regulated production: 2D/3D medical imaging, FDA 510(k) submissions, false-positive and operating-point control for AI used by 50,000+ clinicians in 23,000+ practices.

PROFESSIONAL EXPERIENCE

2025 — Senior ML Scientist at Pearl AI, *Full remote - US dental AI, FDA-cleared 2D + 3D imaging*

Jan-present

- Sole technical owner of the 3D CBCT pathology program (5 pathologies, 8 anatomical structures) since Feb 2026: from the first pathology model to commercial release in Second Opinion 3D.
- Owner of the 2D caries detection model for intra-oral and panoramic X-rays, used by 50,000+ clinicians daily: reduced false positives at constant recall by re-annotating dismissed cases; shipped caries progression on top.
- Operating-point selection for different clinical users (recall-first vs specificity-first), with per-class PR analysis and per-device robustness checks.
- Production inference on AWS, versioned algorithm packages, MLflow tracking, GitLab CI.
- Regulatory evidence for FDA 510(k)s: validation protocol, locked test set with enforced train/test independence, performance reports, technical file; 3D findings turned into claim-ready 2D documentation.

Stack: Python, PyTorch, MONAI, AWS, Docker, MLflow, FastAPI, GitLab CI

2024 — Lead AI Engineer at Theodo Data & AI, *Paris, France*

Sep-Dec

- 3D brain vessel segmentation for surgical trajectory planning, minimally invasive surgery.

2021-2024 — Founding Deep Learning Engineer at Allisone.ai, *Paris, France*

Oct-Feb

- Built the 2D product from zero as one of the first ML engineers: 5+ clinical features shipped on 2D X-rays, ML pipeline owned end to end; the product behind the €10M seed round (2022).
- Led 3D CBCT R&D and built the MVP: volumetric segmentation and generative models on cone-beam CT; co-authored 2 papers (MICCAI 2023 DGM workshop, medRxiv 2024).

Stack: Python, PyTorch, Git, AWS, Docker, FastAPI, ClearML

2021 — Research Engineer at INRIA (team Epione), *Sophia Antipolis, France*

Apr-Sep

Metal Artefact Reduction on Cone Beam CT scans, under the supervision of Hervé Delingette

Stack: Python, PyTorch, Git

2019-2020 — Data Scientist at Société Générale, Data Lab GTPS, *Paris La Défense*

FX margin analysis, transaction data

PUBLICATIONS

2024 — SISTR: Sinus and Inferior alveolar nerve Segmentation with Targeted Refinement on Cone Beam Computed Tomography images

Laura Misrachi, Emma Covili, Hippolyte Mayard, Christian Alaka, Jérémy Rousseau, Willy Au - medRxiv

2023 — Pre-Training with Diffusion Models for Dental Radiography Segmentation - MICCAI 2023, DGM workshop

Jérémy Rousseau, Christian Alaka, Emma Covili, Hippolyte Mayard, Laura Misrachi, Willy Au - arXiv

TEACHING

2022-2026 — External Lecturer at Université Paris Dauphine-PSL

- Deep Learning for Image Analysis - Master IASD/BDIA - Dauphine - PSL University, *Paris/Tunis*
- Deep Learning - Master BDIA - Dauphine - PSL University, *Tunis*

EDUCATION

2020-2021 — Master's degree, MVA (Mathematics, Vision, Learning) - ENS Paris-Saclay

2019-2020 — Master's in Artificial Intelligence and Data Science (IASD) - PSL (Dauphine PSL, ENS, Mines ParisTech)

2014-2018 — Bachelor's degree in Applied Mathematics (with honours) - Université Paris-Dauphine PSL

LANGUAGES AND COMPUTER SKILLS

Programming	Python, C++, SQL
Frameworks	PyTorch, Scikit-Learn, pytest
Vision	Torchvision, OpenCV, Albumentations
Medical frameworks	MONAI, nnU-Net, TorchIO, NiBabel, Pydicom, SimpleITK; FDA clearance documentation & validation
Tools & infra	Git, GitLab CI, Docker, AWS, MLflow, ClearML, FastAPI
Languages	French (Native), English (Fluent), Italian (Fluent), German (Elementary)